The Journal of Chemical Physics

RESEARCH ARTICLE | NOVEMBER 23 2009

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https://doi.org/10.1063/1.3259841

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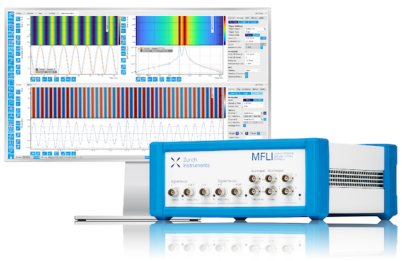
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Construction of theoretical hybrid potential energy curves for LiH($X\ ^1\Sigma^+$)

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(Received 5 August 2009; accepted 19 October 2009; published online 23 November 2009)

Various all-electron and valence-electron potential energy curves for LiH($X\ ^1\Sigma^+$) are compared and assessed. Hybrid potential energy curves are constructed from all-electron potentials at short range and a valence-electron calculation otherwise. This approach provides for the X state of LiH an overall potential curve, which is ionic at equilibrium, and presents an avoided crossing with the excited A state, leading to neutral dissociation products. The classical turning points predicted by these purely theoretical hybrid potentials are compared with those of the experimentally based inverted-perturbation approach (IPA) potentials for both ^{7}LiH and ^{7}LiD. Predicted vibrational energy-level spacings show reasonable ($\lesssim 1\ cm^{-1}$) agreement with the corresponding IPA values. Rotation and vibration-rotation transition energies arising from the most accurate hybrid potential are shown to compare very favorably with recent high-resolution spectroscopic data on ^{7}LiH and ^{7}LiD. © *2009 American Institute of Physics*. [doi:10.1063/1.3259841]

I. INTRODUCTION

Lithium hydride, the simplest stable metallic hydride and the subject of an extensive review in 1993,[1] continues to generate intense theoretical and spectroscopic interest. In particular, it provides a testing ground for new techniques, permits validation of approximate methods, and the existence of various isotopes allows analysis of the breakdown of the Born–Oppenheimer (BO) approximation. Of additional interest are the mutual neutralization of Li^+ and H^- ions,[2,3] the astrophysical applications of line formation in stellar atmospheres,[4] of the radiative association of Li and H,[5,6] and of the potential observation of LiH in the interstellar medium with the recently launched Herschel telescope,[7] the Stark deceleration of a beam of LiH molecules,[8] laser-assisted ultracold LiH formation,[9] and the calculation of the scattering length for the cooling of spin-polarized H by Li.[10] More recently, higher electronic states of LiH have been investigated, theoretically and experimentally, specifically the C (Refs. 11–14) and D (Refs. 15–17) states, where breakdown of the BO approximation is of particular concern.

With regard to advances in our knowledge of the X state, we note the observation of high-resolution spectroscopic data in the millimeter[18,19] and the far- and mid-infrared regions[20] and a recent paper[21] describing direct potential fitting for the X (and A) states. From the theoretical point of view, where LiH provides an ideal system for benchmark, high-quality calculations, potential curves have been determined by full configuration interaction (FCI) (Ref. 22) and multireference coupled-cluster[23] all-electron calculations, as well as valence-electron[24,25] calculations. All-electron coupled-cluster singles and doubles with perturbative triples calculations have been reported by Gadéa and Leininger.[25] While values of various spectroscopic constants and vibrational spacings are provided, unfortunately they do not report the potential curve itself so we have been unable to employ this potential further in our study. It does have an error in the dissociation energy, an order of magnitude smaller than the previous best *ab initio* determination.[22]

The interaction potential of LiH has been included by Geum *et al.*[26] in a series of large-scale configuration-interaction calculations of the interaction of ground state alkali-metal atoms with hydrogen atoms. The equilibrium internuclear separation, harmonic vibrational frequency, and dissociation energy of LiH have been calculated by Smith *et al.*[27] within a set of *ab initio* calculations on the structure and properties of the hydrides of light elements. In addition, all-electron quantum Monte Carlo[28] and valence-electron pseudopotential quantum Monte Carlo[29] calculations have been carried out at equilibrium, as well as explicitly correlated coupled-cluster[30,31] calculations. Recently, there has been a series of non-BO calculations[32] on the ground state of LiH in conjunction with its first vibrational transition.[33]

Theory and experiment are generally compared through examination of vibrational term values and rotational constants, which provide a powerful tool for assessing the accuracy of the attractive region of computed potentials arising from *ab initio* calculations. The presence of an avoided crossing with the excited $A\ ^1\Sigma^+$ potential energy curve (PEC) provides a challenge to theory in that the X state is ionic at equilibrium geometry and is neutral [dissociating to Li(2s) +H(1s)] at large internuclear separations. With only four electrons, FCI can be carried out with a basis set of realistic size.[22] In the case of valence-electron calculations, where the core electrons are taken into account through a pseudopotential, FCI may be carried out using even larger basis sets.[25] In addition, it is possible to compare calculated PECs for the X state with that generated from the experimentally derived inverted-perturbation approach (IPA).[34]

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TABLE I. Comparison of exact atomic and BO energies with the calculated energies from Refs. 22 and 38.

Species	Exact atomic and BO E (E_h)	Ref. 38		Ref. 22	
		E (E_h)	ΔE[a] (cm^{-1})	E (E_h)	ΔE (cm^{-1})
Li	−7.478 060	−7.477 48	127	−7.474 936	685.6
H	−0.500 000	−0.499 996 5	0.8	−0.499 785	47.2
LiH(r_e)	−8.070 549	−8.069 04	331	−8.066 315	929.3
Li^+	−7.279 913	−7.279 36	148	−7.276 950	650.4
H^-	−0.527 751	−0.527 47	53	−0.526 255	328.3
Li+H	−7.978 060	−7.977 48	128	−7.974 721	732.8
Li^++H^-	−7.807 664	−7.806 83	200	−7.803 205	978.7
Spacings	E (cm^{-1})	E (cm^{-1})	ΔE (cm^{-1})	E (cm^{-1})	ΔE (cm^{-1})
D_e	20 298.8	200 95	−204	20 102.6	−196.2
(Li^++H^-)−(Li+H)	37 397.6	374 53	55	37 643.4	245.8
(Li^++H^-)−LiH(r_e)	57 696.4	575 48	−148	57 746.0	+49.6
Li^+−Li	43 488.2	434 82	6	43 452.9	−35.3
H−H^-	6 090.6	6 029	−62	5 809.5	−281.1

[a]ΔE is the difference between the calculated and the exact atomic and BO values.

The main objective of this paper is to assess the accuracy of these various theoretical potential curves for the X state and identify the strengths and weaknesses of each. Direct comparison with experiment is not carried out initially: and a purely theoretical treatment is maintained. The critical region of the potential is in the vicinity of the avoided crossing (around 3.6 Å) between the ground ionic state at smaller values of the internuclear separation, r, and the neutral channel at larger values of r into neutral products, Li(2s)+H. Failure to treat this region accurately normally leads to a well depth which is too shallow, with subsequent errors in the prediction of the higher vibrational bound levels. The valence-electron calculation[25] is known to have an equilibrium bond length that is too short and a well depth that is too deep, suggesting a deficiency in the short-range repulsion, in comparison with that calculated from the all-electron calculations. A simple procedure to rectify this deficiency will be shown to result in hybrid potentials considerably improved over the results of the calculations upon which they are based.

The outline of this paper is as follows. In Sec. II we compare the *ab initio* results at the minimum of the potential and at infinite separation, both for the neutral and ionic limits, in two cases where such information is readily available. Also the various all-electron potentials are compared to the valence-electron potential via difference plots, defining infinite separation of the neutral atoms as the zero of energy. Identification of a separation where the valence and all-electron potentials begin to diverge leads to the construction of hybrid potentials, discussed in Sec. III, based on the all-electron potentials at short range and the valence-electron potential at intermediate to long range, thereby taking advantage of the complementary strengths of the various calculations. These theoretical hybrid potentials are then compared in Sec. IV, firstly with the IPA potentials for ^{7}LiH and ^{7}LiD, for which the classical turning points, vibrational energy levels, and energy-level spacings are tabulated,[1] and secondly, the most accurate hybrid potential is used to determine rotation and rotation-vibration transition energies for comparison with recent experimental high-resolution spectroscopic data[18–20] on ^{7}LiH and ^{7}LiD. This paper ends in Sec. V with some general conclusions.

II. THEORETICAL COMPARISONS

A. Comparison with exact nonrelativistic atomic and BO energies

The existence of a pronounced dipole moment[23,35,36] in the well region of LiH confirms the ionic nature of the bond at these separations, as discussed long ago by Mulliken.[37] Many single-point calculations of the equilibrium ground state of LiH take this ionic feature into account (see, for example, Refs. 30, 38, and 39) and calculate the value of the energy in the ionic (Li^++H^-) limit. In the case of PEC calculations, the valence-electron calculations of Gadéa and coworkers (see, for example, Refs. 25 and 40) take specific note of both ionic and neutral channels, whereas relatively sophisticated all-electron calculations, e.g., Refs. 22, 23, and 35, generally fail to publish the value of their calculated energy at the ionic limit. However, Lundsgaard and Rudolf[22] do publish their energy for the Li^+ ion, and that of the H^- ion may be obtained from their calculated value of the electron affinity of hydrogen. The three limits, namely, equilibrium LiH, Li+H, and Li^++H^-, allow an assessment of the accuracy of the various calculations and may be compared with exact atomic and nonrelativistic BO results.[31] An earlier prediction[38] of the BO energy of equilibrium LiH, r_e = 1.5955 Å, has been improved recently by Cencek and Rychlewski[31] and their result may be combined with the known energies of H, H^-,[41] Li,[42] and Li^+ (Ref. 41) to identify the relevant exact BO limits (see Table I). In view of the paucity of calculations providing information on all desired limits, we concentrate on two calculations, which do so, namely, a single-point calculation[38] and a PEC calculation.[22] The former[38] involves a large Gaussian Type Orbital (GTO) basis in a variational multireference single and double CI calculation based on a complete active space self consistent field (SCF) wavefunction. The latter[22] is a FCI calculation of

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TABLE II. Approximate energies and energy differences from atomic and BO values for calculations A_1 (Ref. 22), A_2 (Ref. 23), A_3 (Ref. 35), and V_v (Ref. 23).

Calculation	E_1(Li+H) (E_h)	ΔE_1[a] (cm^{-1})	E_2(LiH(r_e)) (E_h)	ΔE_2[b] (cm^{-1})	D_e (cm^{-1})	ΔD_e[c] (cm^{-1})	r_e (Å)
A_1[d]	−7.974 721	732.8	−8.066 315	929.3	20 102.6	−196.4	1.5953
A_2[e]	−7.965 181	2826.7	−8.057 090	2953.9	20 171.8	−127.2	1.5980
A_3[f]	−7.972 866	1139.9	−8.063 861	1467.8	19 971.0	−327.8	1.5987
V_v[g]	⋯	⋯	⋯	15.0	20 349.0	+50.0	1.5889

[a]Difference in E_1 from exact atomic energies given in Table I.
[b]Difference in E_2 from exact BO energy given in Table I.
[c]Difference in D_e from exact BO dissociation energy given in Table I.
[d]Reference 22.
[e]Reference 23.
[f]Reference 35.
[g]Reference 25.

the ground state potential energy with a smaller GTO basis. Full details of these calculations are contained in the original papers.

Table I compares the results of these two calculations with the exact nonrelativistic atomic and BO results for the various neutral and ionic species relevant to $X\ ^1\Sigma^+$. Energy differences are quoted in wavenumbers to facilitate comparisons with values reflecting the dissociation of LiH into neutral and ionic fragments. Although the single-point calculation[38] is superior energetically for the absolute values of all species, including the electron affinity of H and the ionization potential of Li, the situation is less clear-cut in relation to the *relative* energies. Specifically, the predicted values of well depth, while very similar, differ by around 200 cm^{-1} from the exact well depth[1] of 20 299 cm^{-1} (see Table I). Although the energy gap between neutral and ionic limits is predicted more accurately by the single-point calculation[38] (thereby reflecting the *relative* accuracy of the separate neutral and ionic limits), the energy required to dissociate equilibrium LiH into ionic species is more accurately predicted by the PEC calculation,[22] with an error of ≈50 cm^{-1} in ≈57 700 cm^{-1}. However, examination of Table I shows that this accuracy is a result of cancellation of errors. We may conclude that despite the amount of effort required in these calculations, the errors which remain, particularly in regard to the well depth, and hence to the experimentally observed dissociation energy, remain significant.

The energy required to dissociate LiH at its equilibrium separation to its ionic limit is not directly involved in the ground state PEC. However, an accurate representation of the ionic behavior in the well region and extending toward the avoided crossing with the A state is necessary in order to predict accurately the position of the avoided crossing and also influences the prediction of the well depth of the PEC itself. The importance of the energy gap between the neutral and ionic separated-atom limits has been emphasized previously by Gadéa and co-workers, e.g., Ref. 40, in their valence-electron calculations. We note that a *balanced* treatment of the equilibrium region, in addition to that of the neutral and ionic limits, is required for an accurate prediction of the well depth.

Next we will compare the predicted equilibrium energies, bond lengths, and dissociation energies from various all-electron[22,23,35] calculations with those from a valence-electron[25] calculation and with the exact BO values to assess the strengths and weaknesses of the various calculations.

B. Comparison of selected PEC calculations

The all-electron PEC calculations considered here have either been widely used[35] or have been shown to attain relatively high accuracy in their predictions of low-lying vibrational-level separations[22,23] in comparison with experiment.[1] We also consider the X-state potential included in the recent extensive valence-electron calculation of ground and excited states of LiH by Gadéa and Leininger.[25] Table II displays for each calculation the absolute and relative energies (in comparison with the atomic and BO results) for Li+H, LiH, and the well depth, D_e. Also tabulated are the bond lengths predicted by the various calculations.

The valence-electron calculation is characterized by an exact ionization energy for Li due to the parametrization of the particular model and core-polarization potentials used in the calculation.[43] The use of a very large basis ensures that the energy of H is essentially exact and the remaining error in H^- is around 15 cm^{-1}.[25] Hence, the error in the energy gap between the neutral and ionic limits is essentially that of the electron affinity of H, namely, around 15 cm^{-1}. As regards the all-electron calculations, we note from Table II that the most accurate prediction of equilibrium bond length is that of Lundsgaard and Rudolf,[22] whereas the most accurate prediction of the well depth is that of Li and Paldus,[23] although their absolute error remains around 130 cm^{-1}.

Overall the most accurate prediction of the well depth is that of the valence-electron calculation,[25] with a value which is too large by some 50 cm^{-1}. However, this is accompanied by a value of the equilibrium bond length which is somewhat shorter than those of the all-electron calculations and of the exact BO value. Gadéa and Leininger[25] suggested that this reflects an inadequacy in repulsion arising from the core-polarization potential used in their calculations. We would add that since the equilibrium ground state has a pronounced dipolar character, the Li^+ core-polarization potential, which

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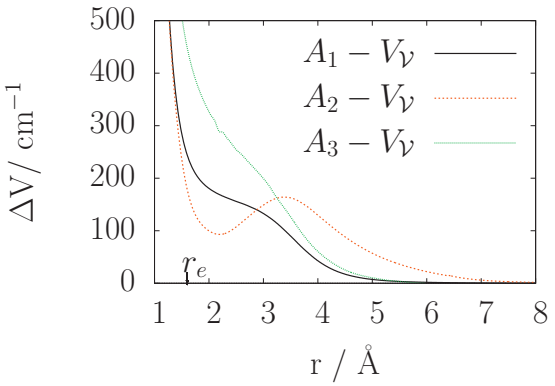


FIG. 1. Difference potentials, A_n-V_ν, as a function of internuclear separation r. The equilibrium separation is indicated with an arrow.

was designed to reflect the behavior of the core within Li itself, is unable to respond appropriately to the presence of H^-.

In order to compare the various all-electron potentials with the valence-electron potential, we use the facility of the Le Roy programme LEVEL (Ref. 44) to mterpolate potentials from the tabulated *ab initio* data. Because the differences are relatively small, we present in Fig. 1 the deviations A_1-V_ν,[22] A_2-V_ν,[23] and A_3-V_ν (Ref. 35) of the *ab initio* potentials, A_n, from the valence, V_ν, potential.[25]

It should be noted that the application of the programme LEVEL to the published data of Partridge and Langhoff[35] produced 26 bound levels of ^{7}LiH rather than the quoted value of 23. A closer examination of their data showed an excessively large energy gap in the region between $15a_0$ and $17.5a_0$. After eliminating this feature by omitting their final three data points, LEVEL then produced a potential with 23 bound levels and a well depth of 19 898.0 cm^{-1}. This particular potential was then used to generate the difference potential A_3-V_ν displayed in Fig. 1.

It is clear from Fig. 1 that the three all-electron potentials deviate from the valence-electron potential in the region 2–7 Å, where the valence potential is expected to be most accurate; both A_1 and A_3 show an increasing deviation from V_ν as r is decreased through the region (between 5 and 2 Å) characterized by the avoided crossing. However, the deviation shown by A_2 displays a maximum around 3.6 Å (the avoided-crossing separation) and then decreases rapidly until the minimum at around 2.2 Å. All three potentials deviate strongly from the valence potential at values of r shorter than 2.2 Å, and a slight "shoulder" may be discerned in that region in both A_1 and A_3. Given that the valence calculation shows too deep a well and too short an equilibrium bond length, we may conclude that the deviations observed for $r<2.2$ Å may be attributed to a deficiency in repulsion within the valence calculation itself.

The A_2 calculation[23] merits further comment, having involved a sophisticated multireference coupled-cluster approach designed to take bond breaking into account. Although the ionic limit is not quoted[23] we may assume from the behavior of the difference potential A_2-V_ν within the region of the avoided crossing that the ionic limit has been described more accurately than the neutral limit. Note that the well depth was also described more accurately by A_2 than by A_1 or A_3 (see Table II). As noted above, the region between 5 and 2.2 Å includes the avoided-crossing region between the X and A states, such that the wavefunction becomes increasingly ionic as r decreases, as reflected in the published variation in dipole moment (see, for example, Refs. 22, 35, and 36). The deficiency in repulsion in the short-range behavior of the valence calculation suggests that the core-polarization potential of Li may be unable to deal adequately with the ionic nature of the LiH bond at short range. This inadequacy in repulsion is reflected in a potential well which is too deep and has a minimum which occurs at too short a value of r.

As we will show in Sec. III, these various deficiencies may be avoided to generate more accurate hybrid potential curves for LiH($X\ ^1\Sigma^+$), which take into account the features of an ionic bond which dissociates into neutral products as a consequence of an avoided crossing with an excited state.

III. CONSTRUCTION OF *AB INITIO* HYBRID PECs

As discussed in connection with Fig. 1, the valence-electron calculation provides a reasonably accurate representation of the true potential in the intermediate-range region in the vicinity of the avoided crossing and also at long range, having represented accurately both ionic and neutral limits. However, the strong deviations between the all-electron calculations, A_n, and the valence-electron calculation, V_ν, at shorter range may be attributed to a repulsive deficiency in the valence calculation itself. A closer examination of Fig. 1 shows that this repulsive deficiency will come into play at the minimum at 2.2 Å in the case of A_2; a slight shoulder in the gradient of the difference potential is visible in the same region for A_1 and A_3.

This suggests that the repulsive deficiency in the valence potential could be corrected via the addition of a repulsive correction, determined from an all-electron calculation as follows. Defining $\Delta V_n(r)=V_{A_n}(r)-V_\nu(r)$, and denoting by r^* the value of r at which the repulsive correction takes effect, then the repulsive correction, $V_{\mathrm{rep},n}$, will be given by

$$V_{\mathrm{rep},n}(r)=\Delta V_n(r)-\Delta V_n(r^*), \quad r\leqslant r^*,$$

$$=0, \quad r\geqslant r^*.$$

Hybrid *ab initio* potentials $V_{H_n}(r)$ are then defined by

$$V_{H_n}(r)=V_\nu(r)+V_{\mathrm{rep},n}(r),$$

such that

$$V_{H_n}(r)=V_\nu(r), \quad r\geqslant r^*$$

$$=V_{A_n}(r)-\Delta V_n(r^*), \quad r\leqslant r^*.$$

This construction is equivalent to combining the all-electron and valence-electron potentials at $r=r^*$ such that $V_{H_n}(r)$ corresponds to the all-electron potential, lowered by an amount $\Delta V_n(r^*)$ for $r\leqslant r^*$ and corresponds to the valence-electron potential for $r\geqslant r^*$. The well depths predicted by these hybrid potentials, V_{H_n}, derived from the corresponding

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TABLE III. Comparison of the calculated well depths and bond lengths of hybrid potentials with exact BO values.

Calculation	D_e (cm^{-1})	ΔD_e (cm^{-1})[a]	r_e (Å)
Exact BO	20 298.8	…	1.595
V_{H_1}	20 271.0	−27.8	1.5953
V_{H_2}	20 260.7	−38.1	1.5980
V_{H_3}	20 186.2	−112.6	1.5987

[a]Difference from exact BO value.

all-electron calculations A_n, are shown in Table III, where the relevant values of D_e show a considerable improvement over the parent calculations—with errors of 28, 38, and 113 cm^{-1}, respectively. This method of construction ensures that the equilibrium bond lengths remain unchanged between the parent and the hybrid potentials. It should be emphasized that these hybrid potentials have been constructed through a combination of purely theoretical calculations and explicit involvement of experimental data has not been necessary.

The significance of these hybrid potentials from the viewpoint of all-electron and valence-electron calculations merits comment. A high-quality valence calculation has been augmented by a short-range repulsive correction, derived through comparison with the results of all-electron calculations. The behavior of the valence potential at long range is assured through use of a sufficiently large basis set, capable of providing comparable relative (and, here, absolute) accuracy for both the neutral and ionic dissociation limits. In the case of all-electron calculations, we focus on the FCI calculation A_1,[22] for which an accurate representation of the potential around equilibrium is provided and for which the relative accuracies of the neutral and ionic limits are known (see Table I). The hybrid potential V_{H_1} acquires the shape of potential A_1 in the region $r<r^*$, but is lowered in energy by around 170 cm^{-1} in order to ensure continuity with the valence-electron potential at $r=r^*$. The lowering is constant across this range of values of r, where the wavefunction remains ionic in character, suggesting that an improved basis would still be required. Table I shows that the difference in accuracy between the ionic and neutral limits for calculation A_1 is 240 cm^{-1}, reflecting, in particular, a poor description of H^- relative to that of H. For calculation A_1, the energy difference between the potential minimum and the separated-ion limit is described to within 50 cm^{-1} in a total of 58 000 cm^{-1}, and an improved basis would affect both the equilibrium region and the ionic limit. Thus, any lowering in energy of the potential in the region $r<r^*$ would be accompanied by a lowering in the value of the energy at the ionic limit, and thus an improvement in the relative energy gap between the ionic and neutral dissociation limits. Hence, the origin of the errors in well depth found in all-electron calculations may be attributed to the use of a basis set which is unable to describe the atomic and ionic limits to similar accuracy. That such a requirement is necessary arises from the presence in the ground-state potential of an avoided crossing involving wavefunctions of ionic and neutral character, such that the neutral and ionic dissociation limits both have a role to play, the former explicitly, the latter implicitly.

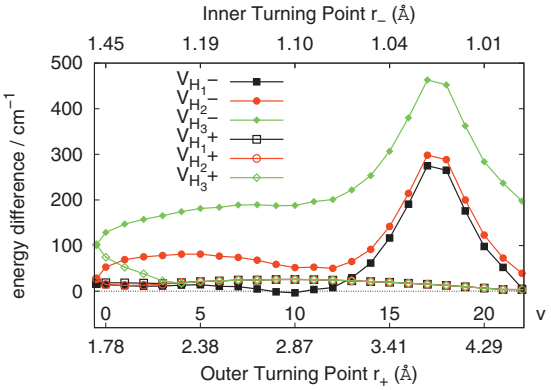


FIG. 2. Energy differences between hybrid potential curves V_{H_n}, and the IPA potential curve at inner, r_-, and outer, r_+, turning points of the IPA potential. The corresponding vibrational quantum numbers are also indicated.

IV. COMPARISON OF HYBRID POTENTIALS WITH EXPERIMENTALLY DERIVED DATA

A. Comparison with IPA potential

The IPA, proposed by Kosman and Hinze,[45] developed by Vidal and Scheingraber,[46] and applied to LiH by Vidal and Stwalley and co-workers,[34,47] provides a convenient format for the representation of spectroscopic data. The IPA potential curve is based on an analysis of spectroscopic measurements and includes adiabatic corrections, thereby yielding a slightly different potential curve for each isotopomer. The IPA curves for the various isotopomers of LiH($X\ ^1\Sigma^+$) are included in the extensive review of LiH by Stwalley and Zemke.[1] Although our theoretical calculations do not include adiabatic corrections, we will compare our approximate BO curves with those for two isotopes, ^{7}LiH and ^{7}LiD, where the well depths are known[1] to be 20 288 cm^{-1} for ^{7}LiH, 20 293 cm^{-1} for ^{7}LiD, in comparison to the BO (infinite mass) value of 20 299 cm^{-1}.

Figure 2 shows for ^{7}LiH the deviations between the three hybrid potentials V_{H_n}, and the IPA potential at the inner, r_-, and outer, r_+, turning points of the vibrational levels of the IPA potential. The energy zero was chosen as the separated-atom limit in each case. Note that the hybrid potentials are equivalent, by construction, for $r \geqslant r^*=2.2$ Å and hence the errors are common for almost all the outer turning points. Deviations from the IPA do not exceed 25 cm^{-1} in either V_{H_1} or V_{H_2} in the attractive portion of the well, whereas V_{H_3} is slightly less accurate, corresponding to an error in well depth of around 100 cm^{-1} (see Sec. III). In the repulsive region, V_{H_1} provides a more accurate representation than does V_{H_2}, reflecting the fact that the values of D_e and r_e are described more accurately by V_{H_1} than by V_{H_2}. Of particular significance is the deviation of all three hybrid potentials from the IPA potentials at very short range ($1.0 \leqslant r$ (Å) $\leqslant 1.05$); this applies equally to ^{7}LiH and ^{7}LiD (not shown). There would appear to be an unphysical "bulge" in the IPA potentials in this narrow region; however, as noted by Chan *et al.*,[34] the inner turning points of the highest vibrational levels of the IPA potentials for the three isotopomers are outside the experimentally defined regions and not as significant as the rest of the data.

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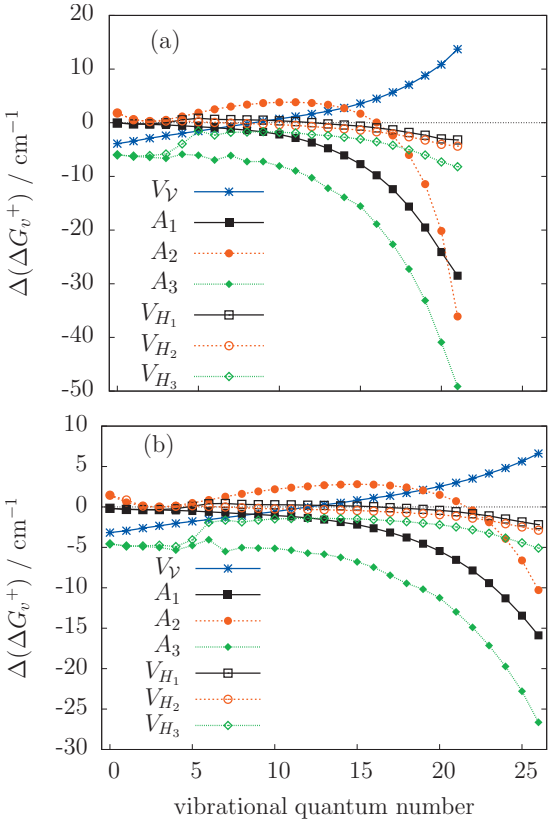


FIG. 3. Deviations, $\Delta(\Delta G_v^+)$, of the predicted vibrational spacings ΔG_v^+ from the experimental values for various calculations (see text) as a function of vibrational quantum number for (a) ^{7}LiH and (b) ^{7}LiD.

Figure 3 shows the deviation of the predicted vibrational spacings, $\Delta G_v^+ = G(v+1) - G(v)$, from the experimental values for ^{7}LiH and ^{7}LiD. We have included results from the three all-electron potentials A_n and the valence-electron potential V_v, in addition to those of the three hybrid potentials V_{H_n}. The improved accuracy of the predictions of the hybrid potentials (particularly V_{H_1} and V_{H_2}) is striking.

For these hybrid potentials, the slight variations at high values of vibrational quantum number show the effects of slightly different well depths and short-range repulsion since the long-range attractive regions are identical. We show in Fig. 4 the vibrational energy levels, G_v, predicted by the most accurate hybrid (by comparison with exact BO values

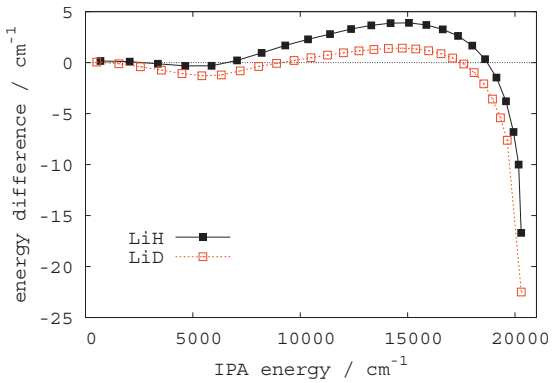


FIG. 4. Deviations of vibrational levels of potential V_{H_1} from those of IPA for ^{7}LiH and ^{7}LiD as a function of the IPA energy.

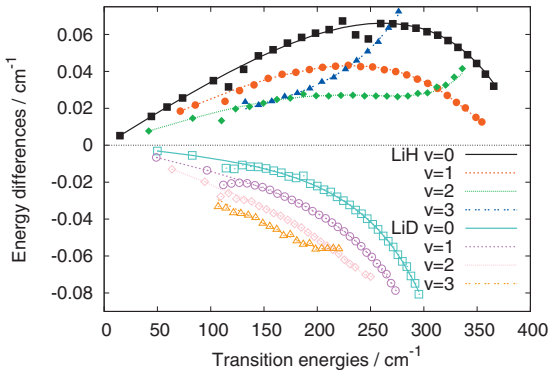


FIG. 5. Rotational transition energy differences (that of V_{H_1}—observed) as a function of the observed (Ref. 19) transition energy (points) for ^{7}LiH and ^{7}LiD. Smooth curves show differences between theory and values calculated using the Dunham spectroscopic constants derived from the observations.

of D_e and r_e) potential V_{H_1} and compared with IPA results for ^{7}LiH. The energy zero is chosen as the minimum of the potential in each case. We note that the potential V_{H_1} supports 24 bound vibrational levels for ^{7}LiH and 32 for ^{7}LiD. The value for ^{7}LiH follows those of other theoretical calculations, e.g., Refs. 22, 23, and 25, whereas the value for ^{7}LiD, while consistent with the calculations considered here, has not been reported previously in general, although the predicted number of bound levels differs more widely from the IPA result than for the case of ^{7}LiH.

B. Comparison with experimental rotational and IR spectroscopic data

In the years following the extensive review on LiH by Stwalley and Zemke,[1] higher resolution spectroscopic data

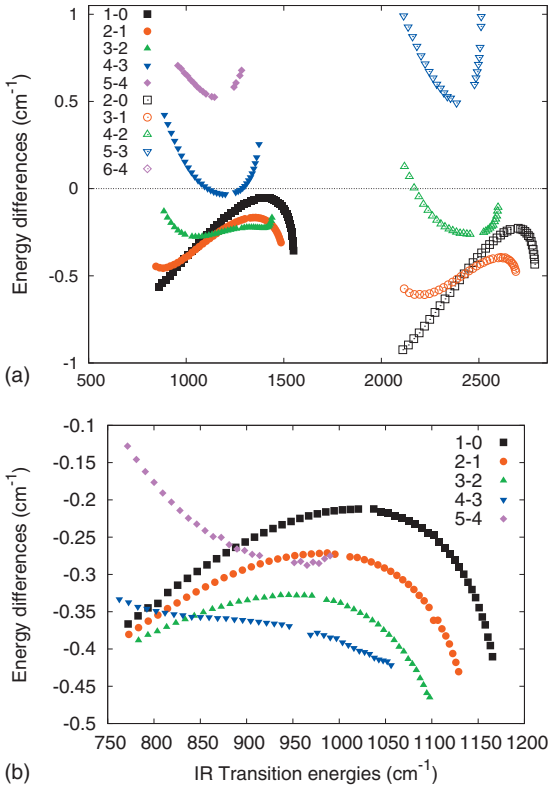


FIG. 6. Infrared transition energy differences (V_{H_1}—observed) as a function of the transition energy for (a) ^{7}LiH and (b) ^{7}LiD.

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TABLE IV. Comparison with experimental values (Ref. 20) of selected ^{7}LiH IR transition energies calculated by Li and Paldus (Ref. 23) and by using the hybrid potential V_{H_1}.

			Differences		
Band	Line	Expt (cm^{-1})	Q[a] (cm^{-1})	(5)[b] (cm^{-1})	V_{H_1} (cm^{-1})
(1,0)	P(28)	858.09	3.5	0.2	−0.57
	P(1)	1344.90	1.9	1.2	−0.07
	R(0)	1374.09	1.8	1.1	−0.06
	R(22)	1550.07	2.5	0.1	−0.36
(2,1)	P(27)	844.92	2.9	1.2	−0.45
	P(1)	1300.47	0.6	0.4	−0.17
	R(0)	1328.82	0.6	0.4	−0.17
	R(17)	1485.86	1.1	0.4	−0.31
(3,2)	P(23)	885.71	2.4	1.8	−0.13
	P(1)	1256.94	0.3	0.5	−0.23
	R(0)	1284.47	0.2	0.5	−0.23
	R(18)	1439.45	1.2	1.2	−0.17
(4,3)	P(21)	888.18	2.4	2.1	0.42
	P(2)	1200.31	0.6	1.0	−0.03
	R(1)	1253.69	0.5	1.0	−0.02
	R(14)	1373.15	1.3	1.5	0.26
(5,4)	P(15)	957.14	2.1	1.9	0.71
	P(3)	1144.87	1.2	1.4	0.52
	R(4)	1244.88	1.2	1.4	0.58
	R(8)	1283.79	1.5	1.6	0.68
(2,0)	P(25)	2109.94	5.8	1.2	−0.92
	P(1)	2659.75	2.4	1.6	−0.23
	R(1)	2701.63	2.5	1.5	−0.23
	R(13)	2785.84	3.5	1.2	−0.43
(3,1)	P(22)	2116.89	4.2	2.5	−0.57
	P(1)	2571.38	0.9	0.9	−0.40
	R(0)	2599.32	0.9	0.8	−0.40
	R(11)	2688.97	1.5	1.1	−0.48
(4,2)	P(19)	2118.45	3.7	3.3	0.13
	P(3)	2454.45	0.8	1.5	−0.26
	R(1)	2524.19	0.7	1.5	−0.25
	R(11)	2598.05	1.6	2.1	−0.11
(5,3)	P(16)	2114.23	3.8	3.6	0.99
	P(2)	2385.37	1.7	2.4	0.49
	R(5)	2477.14	2.0	2.5	0.60
	R(13)	2511.61	3.2	3.3	0.99
(6,4)	P(11)	2145.40	4.0	3.6	1.46
	P(5)	2255.91	3.3	3.2	1.39

[a]Using cc-pVQZ calculations of Ref. 23.
[b]Using cc-pV5Z calculations of Ref. 23.

have been observed in the far-infrared and midinfrared regions,[20] complementing measurements in the millimeter region.[18,19] A large number of pure rotational and rovibrational lines have been measured for the various isotopomers of LiH and LiD. In the following, we calculate pure rotational and rovibrational transitions for the isotopomers ^{7}LiH and ^{7}LiD using the most accurate hybrid potential function, V_{H_1}, for comparison with the experimental observations. As the change in reduced mass on replacing ^{7}Li by ^{6}Li is relatively small, we have not discussed results for ^{6}LiH and ^{6}LiD.

Figure 5 shows the dependence of the pure rotational transition energy differences (calculated for V_{H_1} minus the observed[19,20]) on transition energy for both ^{7}LiH and ^{7}LiD. The tabulated data of Dulick *et al.*[20] were accompanied by residual values (in units of 0.0001 cm^{-1}), representing the difference between the observed and the calculated (via fits from the Dunham spectroscopic constants derived from all their available data) values. The maximum value of these residuals for the pure rotational transitions was around 0.005 cm^{-1}. The relatively smooth curves shown in Fig. 5 are based on the incorporation of these residuals—in effect, we are corroborating the slight discrepancies in the experimental data.

In the case of the IR transitions, plotted in Fig. 6 (with $\Delta v=1$ and 2 for ^{7}LiH and $\Delta v=1$ for ^{7}LiD), the energy differences from experiment[20] reflect errors in vibrational as well as rotational separations and the tabulated data were

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used directly, since the IR transition energy differences were much larger, up to 1 cm^{-1}, than the experimental uncertainties.

The paper by Li and Paldus[23] included comparison between the results of their quartic excitation coupled-cluster calculations (for which the PEC was given) and selected IR transition energies of Dulick *et al.*[20] They also included the corresponding results from their quintet excitation calculations. We show in Table IV for their selection of IR transitions, but restricted to ^{7}LiH for convenience, comparison between our V_{H_1} results, those of Li and Paldus,[23] and experiment.[20] It is clear that our results from the V_{H_1} potential are more accurate in all cases and provide a better representation of these experimental data.

V. CONCLUDING REMARKS

In this comparative study of various published PECs[22,23,25,35] for the X state of LiH we have compared, where data were available, the equilibrium energy and the neutral and ionic dissociation limits with the exact atomic and BO values. If a reasonably accurate representation of the depth of the potential well is to be achieved, an accurate representation of the avoided-crossing region between the X and A states is necessary and this requires a balanced treatment of the neutral and ionic dissociation limits, as well as of the equilibrium region. A closer examination of the relative merits of various all-electron calculations with respect to a recent accurate valence-electron calculation showed that it was possible to combine the best features of each to produce hybrid potentials displaying a more accurate overall behavior. No fitting to experimental data was involved.

These *ab initio* hybrid potentials were then compared with the experimentally derived IPA potentials for ^{7}LiH and ^{7}LiD, where a slight discrepancy in the IPA potentials was noted at very short range (as discussed in Sec. IV A). Comparison of the most accurate hybrid potential (as assessed by the accuracy of the well depth and equilibrium bond length in comparison with the exact BO values) with recent high-resolution spectroscopic data shows close agreement with experiment. Pure rotational transitions agree to within 0.1 cm^{-1} and rovibrational transitions to within around 1 cm^{-1} for both ^{7}LiH and ^{7}LiD. Although the number of predicted bound vibrational levels for ^{7}LiH is 24 (similar to previous calculations), our predicted value for ^{7}LiD is 32, exceeding the observed maximum value (28) quoted for the IPA potential.

The present work confirms the importance of providing a balanced overall calculation in a case where an avoided crossing occurs between an ionic state at equilibrium and a neutral separated-atom limit. The accurate characterization of the avoided crossing is central to an accurate prediction of the depth of the potential well. This characterization requires the use of a basis set capable of determining to comparable accuracy both the neutral and ionic dissociation limits.

Similar requirements are likely to apply to calculations for the other alkali hydrides.

ACKNOWLEDGMENTS

We wish to thank Professor Xavier Gadéa for many helpful discussions. A.S.D. thanks Mr. O. J. Bennett for preliminary calculations.

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